SHIVASELVAM S

Aspiring Software Engineer | Generative AI Applications | LLM Agents and Tool Calling | Voice and Multimodal | Python, TypeScript
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PROFESSIONAL SUMMARY

Computer Science and Business Systems undergraduate (CGPA 7.92/10, graduating 2027) who builds LLM applications end to end. Built a voice agent on the Gemini Live API that calls tools through a relay and stages risky actions behind spoken confirmation; the AI layer of a live Discord product that routes five task types across Gemini and OpenAI-compatible providers with web grounding, conversation memory and token budgets; and a multi-retailer pipeline that forces model output into a strict JSON Schema. Fluent in English; works in Python and TypeScript.

TECHNICAL SKILLS

Generative AI: LLM application design, agents with tool calling and confirmation gates, structured output with JSON Schema, prompt engineering, per-task model routing, web-grounded answering, token budgeting
AI Platforms and APIs: Gemini API (Live, function calling), Ollama (Qwen2.5), OpenAI-compatible APIs (Groq), ONNX Runtime, Whisper, Piper TTS, n8n, ComfyUI image pipelines
Programming: Python, TypeScript, JavaScript, Java, C, C++, Bash
Cloud and Platform: Supabase (PostgreSQL, Auth), Netlify, Docker, GitHub Actions, Linux, WebSockets, REST APIs
Data and ML: Pandas, scikit-learn, Jupyter, classification models
Languages Known: English (Fluent), Tamil (Fluent), Japanese (Elementary)

PROJECTS

Nexa - Voice AI Agent on the Gemini Live API **2026**
Stack: Python, Gemini Live API, WebSockets, Ollama (Qwen2.5), faster-whisper, Piper TTS, ONNX Runtime

- Built a voice agent that streams 16 kHz microphone audio through a WebSocket relay to the Gemini Live API, declares its tools as function declarations, and intercepts every tool call server-side so the API key never reaches the client.
- Put the guardrails outside the model: safe tools (open an app, web search, screenshot) run on call, while consequential ones (typing, key presses, closing a window) are staged and run only after a spoken confirmation, verified end to end against the live API.
- Built a second, fully offline backend behind the same message shape (Whisper, a local tool-calling LLM and Piper TTS on a 4 GB GPU) and cut end-of-speech-to-first-audio latency for commands from 4.5 s to about 0.6 s by profiling every stage.

Xian Ni - AI Layer of a Live Discord Product **2026**
Stack: TypeScript, Node.js, Gemini API, OpenAI-compatible APIs (Groq), SearXNG, SQLite, GitHub Actions

- Built the AI layer behind a 94-command bot: a provider chain across Gemini and OpenAI-compatible endpoints with per-task model routing (quick, chat, reasoning, long context, search) so each command gets a model matched to its latency and context needs.
- Grounded answers that need current information in web results through SearXNG, degrading to an ungrounded answer when a public instance rate-limits, and kept short per-channel conversation memory in process rather than persisting what people say.
- Held cost and abuse in check with a per-user hourly cap and a bot-wide monthly token budget, and gated every change in CI with type checking, linting and 15 test suites that fake each network call.

AI Price Comparison Engine - Structured-Output Pipeline **2026**
github.com/SHIVSSV1269/n8n-workflows-for-Automation | Stack: n8n, LLM API, JSON Schema, JavaScript, REST APIs

- Built a 13-node workflow that fans one request across five retailers and returns a single ranked recommendation: a per-site stage maps messy page text into a strict JSON Schema, then a judging stage compares like-for-like variants and flags suspicious pricing.
- Constrained prompts to return null rather than estimate a price, and handled CAPTCHA walls and JavaScript-only pages with browser headers, a rendering proxy and block detection, so the workflow degrades to a partial comparison instead of failing.

Self-Healing Selenium Framework - Bounded, Auditable LLM Automation **2026**
github.com/SHIVSSV1269/self-healing-selenium-framework | Stack: Java, Selenium, TestNG, Maven, Docker, GitHub Actions, LLM API

- Recovered from renamed UI locators through a tiered pipeline (heal cache, offline structural scoring, then an LLM tier), passing 6/6 unmodified tests against a build where every id and class was renamed, with no API key or network.
- Made each automated decision bounded and auditable: a confidence floor, a uniqueness check and a per-test heal budget make a deleted element fail the run, and every heal is logged to JSON and Allure with before and after selectors.

Other Projects

- **Wildfire** (wildfirespreads.com | Next.js, TypeScript, Supabase, Netlify): a live product with sign-in and watchlists that scores category demand hourly from daily Wikipedia readership.
- **Customer Churn Prediction** (Customer-Churn-Prediction | Python, scikit-learn): chose Logistic Regression on ROC-AUC (0.842) over accuracy with 5-fold cross-validation, since ~73% of customers do not churn.

EDUCATION

B.Tech, Computer Science and Business Systems **Expected 2027**
Panimalar Engineering College, Chennai | CGPA: 7.92 / 10.0
Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS
Class XII: Vivekanandha Academy, CBSE, 2023 - 86.00% | Class X: Vivekanandha Academy, CBSE, 2021 - 79.40%

EXPERIENCE

AI and Machine Learning Intern, InternPe **Jul 2025 - Aug 2025**

- Explored machine-learning models and data-analysis tasks on real datasets, working with a team through data preparation, implementation, and testing.

CERTIFICATIONS, ACHIEVEMENTS AND ACTIVITIES

- Presented a research paper, “Guardian Mobile: A Content Filtering Framework Using SafeSearch and Parental Controls”, at ICONIC 2K26 International Conference on Intelligent Computing, March 2026.
- NPTEL (IIT Madras): Python for Data Science, Elite (76%), 2025 | NASSCOM FutureSkills Prime: Data Science for Beginners, Gold (80%)
- 100 Days of Code: The Complete Python Pro Bootcamp | Udemy (Dr. Angela Yu) | Full-Stack Web Development | QSpiders | AI and ML workshop, IIT Madras